



## EXECUTIVE INSIGHTS

# AI Screening for Diabetic Retinopathy: From Validated Technology to Practical Adoption

### Executive summary

Diabetic retinopathy is a significant complication of diabetes in the U.S. The Centers for Disease Control and Prevention's (CDC) Vision and Eye Health Surveillance System estimates that 9.6 million people in the U.S. had diabetic retinopathy in 2021, including 1.84 million with vision-threatening disease. Among people with diabetes, this represented 26.4% and 5.1%, respectively.<sup>1</sup> The at-risk population is substantial: The CDC estimates that 40.1 million people in the U.S. had diagnosed or undiagnosed diabetes in 2023.<sup>2</sup>

Artificial intelligence (AI) screening has a clear clinical rationale. Diabetic eye disease can develop before symptoms are apparent, and the CDC identifies diabetic retinopathy as the leading cause of blindness in working-age adults.<sup>3</sup> In the U.S., four AI-enabled platforms are Food and Drug Administration (FDA)-authorized for diabetic retinopathy screening: LumineticsCore (Digital Diagnostics), EyeArt (Eyenuk), AEYE-DS (AEYE Health) and iPredict-DR (iHealthScreen).<sup>4</sup> The American Diabetes Association's (ADA) 2026 Standards of Care recognize retinal photography with remote reading or FDA-approved AI algorithms as appropriate screening strategies, provided follow-up pathways are in place.<sup>5</sup>

Adoption has not matched technical availability. The first autonomous AI system was FDA-authorized in 2018, and CPT 92229 has provided a reimbursement pathway for autonomous AI diabetic eye exams since January 2021.<sup>6</sup> Yet a U.S. claims-based study found that AI imaging under CPT 92229 was used in only 0.09% of all patients with diabetes between 2021 and 2023, with use essentially flat over that period.<sup>7</sup>

For medtech companies developing AI-enabled diagnostics, the adoption challenge is shifting from proving that AI can identify diabetic retinopathy to proving that AI screening can work reliably in routine care settings. Current uptake is selective, and broader adoption will depend on integration into organized diabetes-care models with sufficient patient volume, diabetes registries or care-gap lists, nonphysician workflow capacity, medical-record integration and eye-care referral partners. This most strongly favors large primary-care networks and integrated delivery systems, while endocrinology practices and payer-supported clinic programs may also be attractive where they have adequate volume and follow-up infrastructure. In these settings, AI's value is not only faster image interpretation; it is also reducing leakage between overdue eye exams, completed screening and appropriate follow-up.

### **Clinical need and technology validation**

The clinical rationale for diabetic retinopathy screening is well established. Early disease may be asymptomatic, while progression can lead to macular edema, bleeding, scarring, retinal detachment and vision loss.<sup>8</sup> Screening is intended to detect disease before vision loss progresses and to identify patients who need eye care evaluation or treatment.

Autonomous AI systems address a specific step in this pathway: image interpretation at the screening stage. These systems can generate a screening output without specialist image interpretation at the point of screening. They do not eliminate the need for clinical oversight, documentation or referral management. AI therefore shifts the initial screening step into diabetes care settings without replacing specialist eye care.

Many patients with diabetes do not complete a separate eye care visit. Reported barriers include transportation, scheduling, childcare, cost, insurance, referral complexity and low perceived risk when symptoms are absent. AI-enabled retinal imaging can reduce some of that friction by generating a rapid result during a diabetes-related encounter; current tools can provide results within about a minute.

### **Adoption remains selective**

Regulatory clearance, guideline recognition and reimbursement have not translated into broad routine use. Reimbursement matters, but it has not been sufficient to drive adoption on its own. CPT 92229 created a defined U.S. coding pathway for point-of-care autonomous retinal analysis in 2021. The reported national average Medicare payment has remained modest, at \$47.06 in 2022, \$45.75 in 2023, \$40.94 in 2024 and \$43.67 in 2025.<sup>9</sup>

That payment must be evaluated against the full cost of implementation, including hardware, software, staff time, training, information technology (IT) integration, documentation and referral management. The low use of CPT 92229 from 2021 to 2023 suggests that having a reimbursement pathway helps but has not been enough on its own to drive adoption.<sup>10</sup>

The gap between availability and adoption reflects the practical realities of implementation. A clinic must determine who identifies eligible patients, who performs image capture, how image quality is managed, how results enter the record, who communicates findings and how follow-up is arranged. These requirements explain why adoption has not scaled even though the technology is available.

### **The most plausible path to scale**

The most relevant near-term adoption pathway is organized diabetes care. Larger primary care networks, endocrinology practices and integrated delivery systems already manage patients with diabetes and may be able to identify patients due for screening. Not all primary care settings are equally suitable. Fragmented or lower-volume practices may struggle to justify the investment or maintain consistent workflows, especially if the model requires new staff or significant physician involvement.

AI screening is most feasible when it can be incorporated into a protocol-driven delegated workflow. That does not necessarily mean hiring incremental staff. In many settings, the stronger model is to use existing medical assistants, nurses, technicians or diabetes care staff, provided there is sufficient patient volume and capacity. If a site must add dedicated head count solely to operate the camera, the economics become more difficult.

Integrated delivery systems may have an advantage because they can coordinate patient identification, documentation, quality reporting and referral management across a broader network. That advantage is structural, not automatic. Implementation still requires workflow redesign and clear accountability.

Endocrinology practices may offer a strong clinical fit because they see concentrated diabetes populations, including higher-risk or more complex patients. Their limitation is reach. Most diabetes care is not delivered exclusively by endocrinologists, so endocrinology is unlikely to be the broadest population-screening channel on its own.

Community health centers and Federally Qualified Health Centers are relevant where access barriers are substantial. These settings have been cited as attractive locations for AI screening because their patients may be less likely to obtain screening through an eye doctor's office. However, the same settings may face staffing, funding, IT and referral constraints.

Their adoption case depends on implementation support and reliable eye care partnerships, not unmet need alone.

Provider organizations have a reason to accelerate AI screening when it can close a measurable diabetic eye-exam gap without adding physician burden. The strongest use cases are likely to be sites with sufficient diabetes volume, nonphysician workflow capacity, IT support and established eye-care referral relationships. In these settings, AI screening can shift the eye-screening step into routine diabetes care, improve completion of overdue exams and reduce leakage between referral and follow-up.

Payers and risk-bearing organizations can accelerate adoption by creating the conditions for providers to implement screening. Their role is typically not to operate the camera, but to identify members overdue for eye exams, steer them to screening sites, support reimbursement or incentives, and track whether gaps are closed. The model is strongest when payer outreach, provider workflow and eye-care referral access are connected. Without that linkage, AI screening risks increasing test volume without reliably improving follow-up.

### Practical requirements for implementation

The main barriers are operational rather than purely clinical. Implementation requires compatible camera hardware, trained operators, image-quality protocols, workflows for ungradable results, electronic health record (EHR) integration, referral processes, staff training and sustainable economics.

IT integration matters because the screening result must become part of the clinical workflow. At minimum, a scalable model needs to identify eligible patients, support ordering, return results to the medical record, document screening completion and trigger follow-up for positive or ungradable results. More integrated implementations can connect the provider order to the camera system and upload results automatically to the EHR; less integrated models may rely on manual worklists, scanned reports and manual referral tracking, which add administrative burden and weaken scalability.

Staffing is also a meaningful constraint. Even if image capture can be learned quickly, image quality depends on consistent execution. Sites with high turnover among support staff may need repeated training and backup coverage to maintain quality. Lower-volume sites may also struggle if staff perform scans too infrequently to remain proficient.

Hardware is another adoption variable. Current autonomous screening workflows rely on digital fundus photography, typically nonmydriatic unless dilation is needed for image quality.

AEYE-DS has received clearance for use with the Optomed Aurora portable handheld camera, which could expand where screening can be performed. Portability may improve access, but smaller devices can create image-quality trade-offs that affect AI performance.<sup>11</sup>

### Follow-up and downstream capacity

Follow-up is the most important constraint. A positive or ungradable screen creates a clinical obligation. The result must be communicated, the referral must be placed, an appointment must be available and the patient must complete the visit. If patients who screen positive cannot access eye care, higher screening volume may not translate into higher treatment rates.

Treatment for diabetic retinopathy and diabetic macular edema is available, but it can be resource-intensive. The ADA's 2026 Standards of Care note that anti-VEGF treatment can reduce vision loss risk in diabetic retinopathy and diabetic macular edema, while panretinal laser photocoagulation remains indicated for high-risk proliferative diabetic retinopathy and some severe nonproliferative cases.<sup>12</sup>

Treatment burden matters because expanded screening may identify more patients who need eye care evaluation and ongoing management. If specialist access, patient adherence or treatment capacity are constrained, screening expansion may improve detection without proportionately improving outcomes.

### Implications for medtech companies developing AI-enabled diagnostics

For medtech companies developing AI-enabled diagnostics, the next phase of competition will be defined by adoption support, not algorithm performance alone. Clinical validation remains necessary, but customers will need confidence that AI screening can be implemented in routine diabetes care settings without creating excessive workflow burden, staffing requirements or referral leakage.

The most attractive customers are likely to be organized provider networks and payer-supported programs with a clear need to close diabetic eye exam gaps. Large primary care groups, integrated delivery systems, endocrinology practices and Medicare Advantage or risk-bearing programs are more likely to have the patient volume, care-gap data and referral relationships needed to support adoption. Smaller or more fragmented practices may still be addressable, but only if the implementation model is simple and the economics are compelling.

To accelerate adoption, medtech companies will need to help providers solve the practical work around the test. That includes identifying eligible patients, selecting camera

configurations, managing image quality, training staff, integrating results into the EHR or care-management platform, supporting reimbursement and tracking referrals. A product that only generates an accurate image read may struggle if the customer still has to solve patient identification, documentation and follow-up independently.

Evidence generation should also extend beyond sensitivity and specificity. Buyers will want to understand screening completion, ungradable image rates, staff time per screen, referral completion, reimbursement capture, total cost of ownership and performance across different care settings. This data will be especially important for payer-supported and value-based care models.

Radiology AI offers a useful lesson. Radiology has been the most mature clinical AI category: a 2025 JAMA Network Open systematic review found that 723 of 950 FDA-authorized AI/machine learning medical devices, or 76%, were radiology devices.<sup>13</sup> However, radiology AI adoption has still required workflow integration, interoperability and operational redesign. A review published in Radiology notes that AI deployment creates new demands on imaging workflows, that custom integrations can create substantial operational and maintenance burden, and that standards-based interoperability is important for scaling AI integration.<sup>14</sup>

Partnership strategy will therefore be critical. AI screening vendors may need to work with provider groups, payers, eye care networks, and EHR or care management platforms to make the model actionable. Eye care specialists remain essential downstream partners, particularly for positive or ungradable screens and treatment of sight-threatening disease. Without reliable referral capacity, broader screening may increase detection without translating into improved care.

The implication is that the strongest medtech companies in AI diabetic-retinopathy screening may not be those with the best algorithm alone. They will be the companies that make AI screening easiest for customers to implement in terms of finding the right patients, producing reliable images, documenting results, supporting reimbursement and helping move patients to follow-up when needed.

## Conclusion

AI screening for diabetic retinopathy has cleared the validation hurdle. FDA-approved tools exist, ADA guidance recognizes AI-enabled screening pathways and reimbursement has been available for several years. Adoption remains limited largely because clinics still have to solve the practical issues around staffing, workflow, integration and follow-up.

The strongest path to scale is not broad device placement. It is integration into care models that can identify eligible patients, complete imaging efficiently, document results, secure reimbursement and connect abnormal findings to eye care follow-up. Organized provider networks and payer-supported programs with clinical partners are the most credible near-term pathways. High-need community settings may also be important, but only where staffing, IT and referral constraints can be addressed.

For medtech companies developing AI-enabled diagnostics, accuracy remains essential, but adoption will depend on more than the algorithm. The strongest solutions will help providers and payers identify overdue patients, complete screening, document results and coordinate follow-up for positive or ungradable findings.

For more information, please **contact us**.

## Endnotes

<sup>1</sup>CDC.gov, "VEHSS Modeled Estimates: Prevalence of Diabetic Retinopathy (DR)." <https://www.cdc.gov/vision-health-data/prevalence-estimates/dr-prevalence.html>

<sup>2</sup>CDC.gov, "National Diabetes Statistics Report." <https://www.cdc.gov/diabetes/php/data-research/index.html>

<sup>3</sup>CDC.gov, "Vision Loss and Diabetes." <https://www.cdc.gov/diabetes/diabetes-complications/diabetes-and-vision-loss.html>

<sup>4</sup>AAO.org, "Can AI Close the Diabetic Retinopathy Screening Gap?" <https://www.aao.org/eyenet/article/artificial-intelligence-for-diabetic-retinopathy>; AccessData.gov, "510(k) Premarket Notification." <https://www.accessdata.fda.gov/scripts/cdrh/cfdocs/cfpmn/pmn.cfm?ID=K253704>

<sup>5</sup>DiabetesJournals.org, "Retinopathy, Neuropathy, and Foot Care: Standards of Care in Diabetes—2026." [https://diabetesjournals.org/care/article/49/Supplement\\_1/S261/163919/12-Retinopathy-Neuropathy-and-Foot-Care-Standards](https://diabetesjournals.org/care/article/49/Supplement_1/S261/163919/12-Retinopathy-Neuropathy-and-Foot-Care-Standards)

<sup>6</sup>DiabetesJournals.org, "Clinical Implementation of Autonomous Artificial Intelligence Systems for Diabetic Eye Exams: Considerations for Success." <https://diabetesjournals.org/clinical/article/42/1/142/153640/Clinical-Implementation-of-Autonomous-Artificial>

<sup>7</sup>JAMANetwork.com, "Use of Artificial Intelligence–Based Detection of Diabetic Retinopathy in the US." <https://jamanetwork.com/journals/jamaophthalmology/fullarticle/2825181>

<sup>8</sup>CDC.gov, "Vision Loss and Diabetes."

<sup>9</sup>AAO.org, "Can AI Close the Diabetic Retinopathy Screening Gap?"

<sup>10</sup>JAMANetwork.com, "Use of Artificial Intelligence–Based Detection of Diabetic Retinopathy in the US."

<sup>11</sup>AAO.org, "Can AI Close the Diabetic Retinopathy Screening Gap?"

<sup>12</sup>DiabetesJournals.org, "Retinopathy, Neuropathy, and Foot Care: Standards of Care in Diabetes—2026."

<sup>13</sup>JAMANetwork.com, "FDA Approval of Artificial Intelligence and Machine Learning Devices in Radiology." <https://jamanetwork.com/journals/jamanetworkopen/fullarticle/2841066>

<sup>14</sup>Pubs.RSNA.org, "Integrating and Adopting AI in the Radiology Workflow: A Primer for Standards and Integrating the Healthcare Enterprise (IHE) Profiles." <https://pubs.rsna.org/doi/10.1148/radiol.232653>

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